

Cristobal Cortés Gómez

Game Developer & Mathematician

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Professional Profile

Cross-platform game developer across desktop, mobile and web, with a focus on game feel and optimization for low-end devices. Advanced English, used daily with U.S.-based clients.

Education

2025: Bachelor's Degree in Mathematics — Universidad de Guadalajara (CUCEI). Final project: "Suitability Analysis of Shallow vs. Deep Neural Networks Using the Perceptron for Classification Problems", built on a self-made neural network in C and CUDA.

Technical Skills

Programming languages: TypeScript/JavaScript, Python, C, C#, GML, GLSL.

Web, frameworks & tools: HTML/CSS, Node.js, React, Redux, Three.js, PixiJS, Firebase.

Game Engines: GameMaker, Unity.

Systems & hardware: Linux (shell scripting), Git; ESP8266, Arduino.

Spoken languages: Spanish (native); English (advanced).

Experience

Jun. 2025–Jun. 2026: Game Developer — *Gnarled Helix LLC (~10-person U.S. studio), remote contract*

- Gameplay, AI, shaders and UI for "Open Sourcerer", a browser-based 2D action game that teaches programming via an in-game code editor, built on the studio's custom engine (TypeScript, Three.js, React, Redux, WASM physics).
- Authored multiple enemy AIs on the engine's behavior-tree framework, built the inventory UI in React and a Three.js renderer that draws the skill tree from nodes authored in Tiled level format, and wrote a custom god-ray shader for a cinematic level.
- Reworked the player's mouse-driven parabolic aiming, deriving the launch solution by hand to cover cases where the quadratic has no real root.

2024–2025: Sole Developer — *Jalapeño Labs, contract; playable online, development paused at contract end*

- Developed the 3D digital adaptation of the board game "Camino a Xibalbá PITZ", a chess-like strategy game inspired by the Mesoamerican ballgame, for browser and mobile (GameMaker, JavaScript, Firebase).
- Built the rule-resolution system and full turn-based online multiplayer with user accounts and match records, backed by Firebase Realtime Database.

2023: Game Developer — *"Wen Sandwich", contract.* Physics-based QWOP-like web3 game. Built a physics-joint character creator tool for the team and an anti-cheat system that records and uploads segments of play (GameMaker, HTTP REST).

Additional Experience

Contracts from 2016 to 2024 span automation, web development, tutoring and game development, alongside completing the Mathematics degree. The most relevant:

2020–2022: Hardware & Software Developer — *Proyecto INTERSEX.* Built an LED face mask and a 3D museum simulator for a social and artistic project (JavaScript, PixiJS, Arduino, ESP8266, Unity).

Selected Independent Projects

The games below were made with the 4-member Axolotl Bros team; all listed on my portfolio.

2019 (paused) "Libélula" Multiplayer fighting game. Custom 2D light/shadow system, parallax backgrounds, and reactive competitive gameplay tuned to details like coyote time and variable-height jumps.

2018 "Sacrifice Your Brothers" (programmer, 2-person code team) Local multiplayer zombie survival game. A* pathfinding and game feel work: frame-skipping screen shake, tuned friction/damping, and per-weapon recoil. 4k downloads on itch.io.

2014–2018 (unreleased) "Gravity Box" Multiplayer experimental arcade game. Multi-gravity movement built on the Extreme Physics library, platformer AI with parabolic shots, and a large gameplay-modifier item set.

2014 "Sweet Valentine's Day" (sole programmer) Arcade parody about love. Mobile adaptability, a Lemmings-like single-pixel collision system, and online high scores; runs on the low-end Android hardware of the time.

2012 "Morbus" (sole programmer) Zombie survival game. Simple lighting system, multi-resolution (hierarchical) A* pathfinding, and a 2D top-down bone animation system.

Global Game Jam (48-hour jams): 2012, 2013, 2014, 2016, 2018, 2019, 2020, 2024; local-site honorable mention in 2019.